

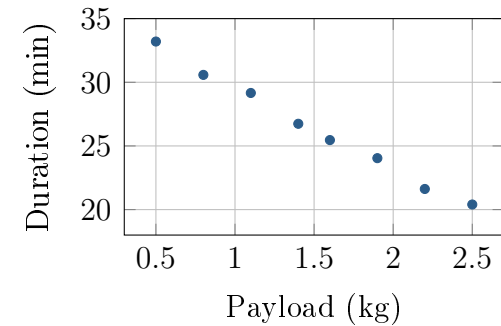
Two Payloads, One Regression Line

Unit 5 · Topic 5.3 · THE PROBLEM

Suppose eight drone tests with payloads from 0.5 to 2.5 kg produced the shown data and the model $\hat{y} = 36.0 - 6.4x$, where x is payload and $\hat{y}$ is predicted flight duration in minutes. A proposed mission carries 3.5 kg and requires 12 minutes of flight.

Ravi: Approve it: the model predicts more than the required 12 minutes, and the fitted data are close to a line.

Mina: Before deciding, compare the support for predictions at 1.8 and 3.5 kg and identify evidence the safety decision still needs.



FIRST TAKE — one sentence before you work the parts

Would you approve now, delay approval, or approve conditionally? Cite one number or plot feature.

- DOK 1** (a) Name the explanatory and response variables, with units.
- DOK 2** (b) Predict flight duration at 1.8 and 3.5 kg. How much does the 3.5-kg prediction exceed the required 12 minutes?
- DOK 3** (c) Would you approve this mission now? Compare the support for the two predictions using the tested payload range and the extra time from (b). Explain what evidence you would want before relying on the 3.5-kg prediction. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.